LUIGUI GALLARDO-BECERRA

BIOINFORMATICIAN | MOLECULAR BIOLOGIST

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CAREER SUMMARY

Computational biologist (Ph.D., Biochemistry) studying how gut microbial communities, their phages, and the peptides they encode shape host health across human, animal, and environmental systems. Experienced in integrating metagenomic, metatranscriptomic, and viromic data to generate biological insight, with 15 peer-reviewed publications and hands-on experience at both the bench and the computer.

RESEARCH EXPERIENCE

Graduate Research Assistant (Bioinformatics & Molecular Biology) | Institute of Biotechnology, UNAM January 2019 - March 2026

Across several interdisciplinary research projects, I contributed to data acquisition, computational analysis, and dissemination of scientific results. My responsibilities included:

- **Project conceptualization:** Assisted in defining research goals, experimental design, analytical workflows, and computational strategies to address multi-omics biological questions.
- **HPC system administration:** Maintained laboratory computational infrastructure, including Linux installation, software configuration, performance optimization, and data security practices.
- **Reproducible workflow engineering:** Built robust, scalable, and reproducible pipelines using Bash, Nextflow, and Snakemake, to automate NGS and metagenomic analyses.
- **Custom tool development:** Implemented specialized algorithms and utilities in Python, R, Bash, and C# to support complex multi-step analyses.
- **Quantitative data visualization:** Designed comprehensive visualizations and figure panels using ggplot2, seaborn, matplotlib, and Jupyter Notebooks.
- **Computational best practices:** Maintained version control through GitHub repositories and incorporated CI/CD testing to ensure workflow reliability.
- **Scientific reporting:** Prepared internal research reports, contributed to manuscript writing, and provided analytical support for high-impact publications.
- **Presentation & outreach:** Delivered oral and poster presentations at national and international conferences and participated in science communication initiatives.

PROFESSIONAL EXPERIENCE

Data Specialist | TEKsystems (contractor at Meta) July 2026 - Present

Supporting Meta's Trust & Safety operations through content evaluation that improves platform integrity and advertiser confidence.

- **Content classification:** Systematically classify and annotate digital content to improve Content Brand Suitability standards across Meta's advertising ecosystem.
- **Content labeling at scale:** Evaluate and label content against established guidelines to ensure brand-safe environments for advertisers.
- **Data quality for machine learning:** Contribute to data quality initiatives that feed machine learning models for automated content moderation and suitability scoring.
- **Policy refinement:** Identify edge cases and provide feedback to cross-functional teams to refine classification frameworks and policy guidelines.

STEM Python Expert | Mercor (remote contractor) April 2026 - Present

Evaluating AI-generated scientific code and reasoning to improve large language model performance in chemistry and biology.

- **AI code evaluation:** Evaluate and benchmark AI-generated Python code for scientific accuracy and correctness in chemistry and biology.
- **Training data for language models:** Contribute to RLHF annotation workflows and build benchmark datasets for STEM reasoning.

EDUCATION

Ph.D. in Biochemistry | National Autonomous University of Mexico (UNAM) | Mexico — January 2019 - March 2026

Master of Science in Biochemistry | National Autonomous University of Mexico (UNAM) | Mexico — August 2016 - January 2019

Bachelor of Science in Biology | University of Guadalajara (UDG) | Guadalajara, Mexico August 2012 - January 2016

TEACHING EXPERIENCE

- Organizer and instructor of the "Bioinfo del Norte" bioinformatics and metagenomics training event, CIAD, Hermosillo, Sonora. September 2024.
- Guest lecturer at Universidad de La Sabana (Chia, Colombia). December 2023.
- Microbiota Symposium from Theory to Clinical Practice, Colegio de Profesionales de la Nutrición de Querétaro y el Bajío. June 2021.
- Bioinformatics and Reproducible Research Workshop, Postgraduate Program in Biological Sciences, UNAM. October 2019.
- Guest lecturer at Center of Genomic Sciences, UNAM. August 2018.

DIRECTED THESES

Bachelor's Thesis: Itzel Abigail Hernández Reyna, Bachelor's in Nutrition (Universidad Autónoma del Estado de Morelos), December 2019. "Impact of bacteriophages associated with childhood obesity on the intestinal metagenome."

TECHNICAL SKILLS

Specialized Bioinformatics Skills: Genome and transcriptome assembly & annotation; differential expression analysis; taxonomic profiling; functional & pathway analysis; ETL automation for multi-omics data

Bioinformatics Tools & Methods: Snakemake, Nextflow, QIIME2, Trinity, SPAdes, Bowtie2, BWA, HISAT2, samtools, Kraken2

NGS & Multi-omics Analysis: RNA-seq, WGS, 16S rRNA profiling, metagenomics, metatranscriptomics, viromics

Molecular Biology Skills: DNA/RNA extraction, PCR/qPCR, gel electrophoresis, sample QC, NGS library preparation, bacterial culture & handling, plasmid preparation, sterile technique (BSL-2), and collaboration with wet-lab teams

Programming Languages: Python, R, Bash, SQL, C#, JavaScript, TypeScript, HTML/CSS

Frameworks & Libraries: React, Django, ASP.NET Core, Shiny

Computing & DevOps: Linux, High-Performance Computing, Docker, Git/GitHub, CI/CD workflows, Conda

Data Science & Visualization: ggplot2, seaborn, matplotlib, Jupyter Notebooks, RStudio

Databases: MySQL, PostgreSQL, MongoDB

LANGUAGES

English – Full professional proficiency

Spanish – Native

PUBLICATIONS

1. Jatuyosporn T, Laohawutthichai P, Cornejo-Granados F, **Gallardo-Becerra L**, Tassanakajon A, Hurtado-Ramírez JM, Ochoa-Leyva A, Krusong K. *De novo* transcriptome assembly and annotation of *Penaeus monodon* hemocytes under WSSV infection and STAT knockdown. Sci Data. 2026.
2. **Gallardo-Becerra L**, Cornejo-Granados F, Bikel S, Arenas I, López-Leal G, Alvarado-Gonzalez C, Sánchez-López F, Manzo R, Corzo G, Espino-Solis GP, Canizales-Quinteros S, Ochoa-Leyva A. Bioactive plasmid- and phage-encoded antimicrobial peptides (AMPs) in the human gut: a metatranscriptome–virome profiling reveals exploratory links to metabolic human diseases. Microb Ecol. 2026;89(1):15. doi:10.1007/s00248-025-02620-2.
3. Cornejo-Granados F, **Gallardo-Becerra L**, Romero-Hidalgo S, Lopez-Zavala AA, Cota-Huízar A, Cervantes-Echeverría M, Sotelo-Mundo RR, Ochoa-Leyva A. Host genome drives the microbiota enrichment of beneficial microbes in shrimp: exploring the hologenome perspective. Animal Microbiome. 2025 May 22;7(1):50.
4. Manzo R, **Gallardo-Becerra L**, Díaz de León-Guerrero S, Villaseñor T, Cornejo-Granados F, Salazar-León J, Ochoa-Leyva A, Pedraza-Alva G, Pérez-Martínez L. Environmental Enrichment Prevents Gut Dysbiosis Progression and Enhances Glucose Metabolism in High-Fat Diet-Induced Obese Mice. International Journal of Molecular Sciences. 2024 Jun 24;25(13):6904.
5. Jatuyosporn T, Laohawutthichai P, Ochoa Romo JP, **Gallardo-Becerra L**, Sánchez Lopez F, Tassanakajon A, Ochoa-Leyva A, Krusong K. White spot syndrome virus impact on the expression of immune genes and gut microbiome of black tiger shrimp Penaeus monodon. Sci Rep. 2023 Jan 18;13(1):996.

6. **Gallardo-Becerra L**, Cervantes-Echeverría M, Cornejo-Granados F, Vazquez-Morado LE, Ochoa-Leyva A. Perspectives in Searching Antimicrobial Peptides (AMPs) Produced by the Microbiota. Microb Ecol. 2023 Dec 1;87(1):8.
7. Chino de la Cruz CM, Cornejo-Granados F, **Gallardo-Becerra L**, Rodríguez-Alegría ME, Ochoa-Leyva A, López Munguía A. Complete genome sequence and characterization of a novel Enterococcus faecium with probiotic potential isolated from the gut of Litopenaeus vannamei. Microbial Genomics. 2023;9(3):000938.
8. Cervantes-Echeverría M, **Gallardo-Becerra L**, Cornejo-Granados F, Ochoa-Leyva A. The Two-Faced Role of crAssphage Subfamilies in Obesity and Metabolic Syndrome: Between Good and Evil. Genes. 2023 Jan;14(1):139.
9. Palomino-Hermosillo YA, Berumen-Varela G, Ochoa-Jiménez VA, Balois-Morales R, Jiménez-Zurita JO, Bautista-Rosales PU, Martínez-González ME, López-Guzmán GG, Cortés-Cruz MA, Guzmán LF, Cornejo-Granados F, **Gallardo-Becerra L**, Ochoa-Leyva A, Alia-Tejacal I. Transcriptome Analysis of Soursop (Annona muricata L.) Fruit under Postharvest Storage Identifies Genes Families Involved in Ripening. Plants (Basel). 2022 Jul 7;11(14):1798.
10. Ochoa-Romo JP, Cornejo-Granados F, Lopez-Zavala AA, Viana MT, Sánchez F, **Gallardo-Becerra L**, Luque-Villegas M, Valdez-López Y, Sotelo-Mundo RR, Cota-Huízar A, López-Munguia A, Ochoa-Leyva A. Agavin induces beneficial microbes in the shrimp microbiota under farming conditions. Sci Rep. 2022 Apr 16;12(1):6392.
11. Bikel S, **Gallardo-Becerra L**, Cornejo-Granados F, Ochoa-Leyva A. Protocol for the isolation, sequencing, and analysis of the gut phageome from human fecal samples. STAR Protocols. 2022 Mar 18;3(1):101170.
12. Bikel S, López-Leal G, Cornejo-Granados F, **Gallardo-Becerra L**, García-López R, Sánchez F, Equihua-Medina E, Ochoa-Romo JP, López-Contreras BE, Canizales-Quinteros S, Hernández-Reyna A, Mendoza-Vargas A, Ochoa-Leyva A. Gut dsDNA virome shows diversity and richness alterations associated with childhood obesity and metabolic syndrome. iScience. 2021 Aug 20;24(8):102900.
13. **Gallardo-Becerra L**, Cornejo-Granados F, García-López R, Valdez-Lara A, Bikel S, Canizales-Quinteros S, López-Contreras BE, Mendoza-Vargas A, Nielsen H, Ochoa-Leyva A. Metatranscriptomic analysis to define the Secrebiome, and 16S rRNA profiling of the gut microbiome in obesity and metabolic syndrome of Mexican children. Microb Cell Fact. 2020 Mar 10;19(1):61.
14. Cornejo-Granados F, **Gallardo-Becerra L**, Leonardo-Reza M, Ochoa-Romo JP, Ochoa-Leyva A. A meta-analysis reveals the environmental and host factors shaping the structure and function of the shrimp microbiota. PeerJ. 2018 Aug 10;6:e5382.
15. Cornejo-Granados F, Lopez-Zavala AA, **Gallardo-Becerra L**, Mendoza-Vargas A, Sánchez F, Vichido R, Brieba LG, Viana MT, Sotelo-Mundo RR, Ochoa-Leyva A. Microbiome of Pacific Whiteleg shrimp reveals differential bacterial community composition between Wild, Aquacultured and AHPND/EMS outbreak conditions. Sci Rep. 2017 Sep 18;7:11783.